

ST JOSEPH'S BOYS' HIGH SCHOOL, BENGALURU
MIDTERM ASSESSMENT 2026
SUBJECT: MATHEMATICS

Std : 12

Date : 28.09.2026

Maximum Marks : 80

Duration : 3 hours

Instructions

- 1) You are allowed 15 minutes to read the paper .
 - 2) Internal choice has been provided in the paper .
 - 3) Answer all questions from each section , preferably in the same order.
 - 4) All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer. This paper has 7 pages.
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SECTION A(20 MARKS)

(i) A relation R is defined on the set of students of a class such that $R = \{(a, b) : \text{student } a \text{ and student } b \text{ have at least one common friend}\}$

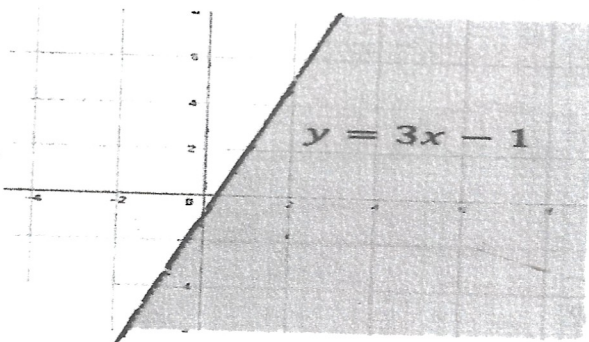
Which of the following is true?

- a) R is an equivalence relation.
- b) R is neither symmetric nor transitive.
- c) R is symmetric but not necessarily transitive.
- d) R is reflexive and transitive but not symmetric.

(ii) The matrix shown is skew symmetric, what is the value of $3x - 2y$.

$$\text{If } A = \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}$$

(iii) State the inequality represented by the shaded region.

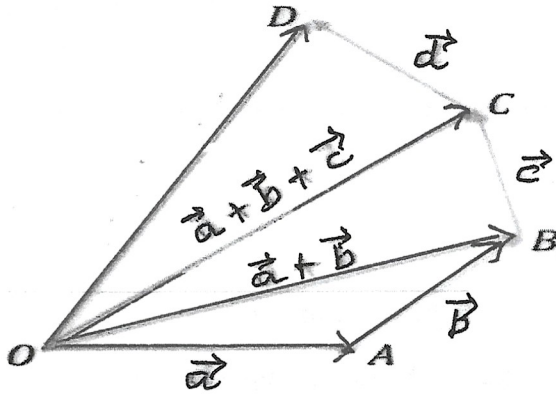


(iv)

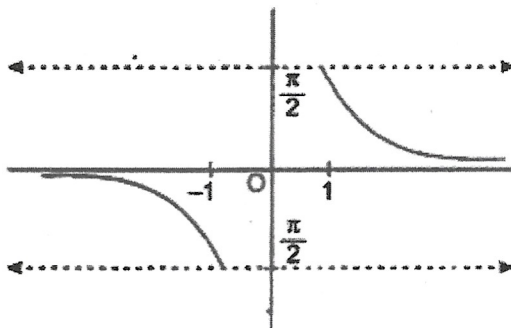
If $P(A) = m$, $P\left(\frac{B}{A}\right) = 3m$, $P\left(\frac{B}{A}\right) = 6m$, then what will be $P\left(\frac{A}{B}\right)$?

(v) In a batch of 25 students at a dance academy, there are 12 boys and 13 girls. Among them, 5 boys and 7 girls are members of the Indian classical dance team. A student is selected at random to lead the team at an international dance festival. If it is known that the selected student is from the Indian classical dance team, find the probability that the student is a girl.

(vi) Find DO vector.

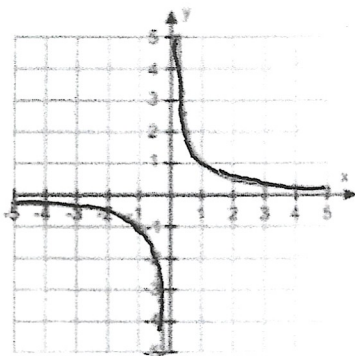


(vii)



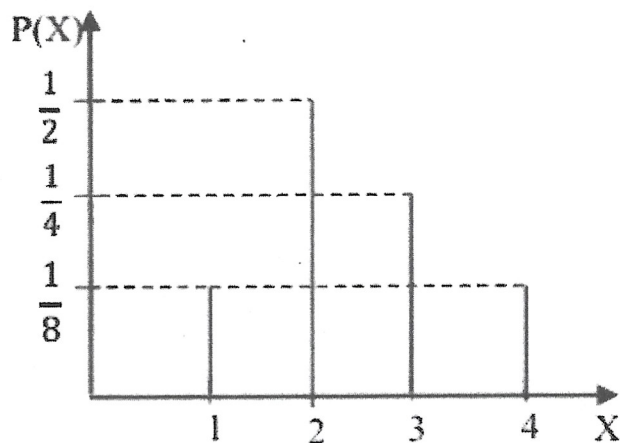
What is the domain of the function displayed in the above graph?

(viii) Find the range of the function shown below?

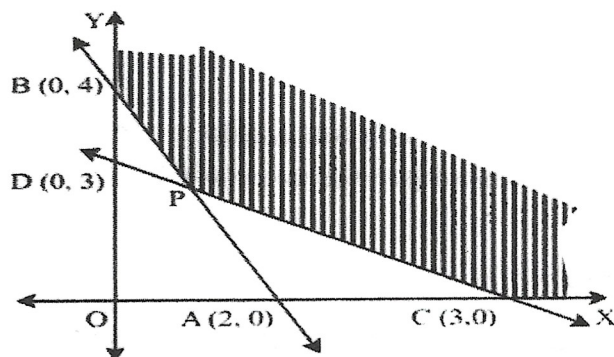


(ix)

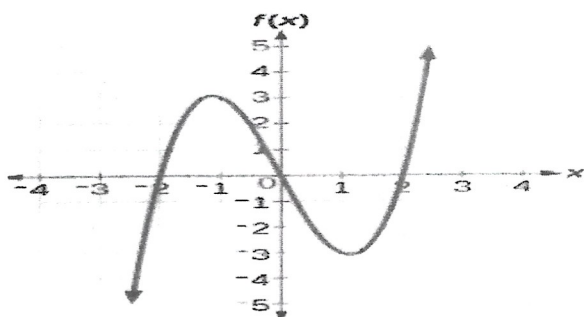
Find the Expectation of the following distribution?



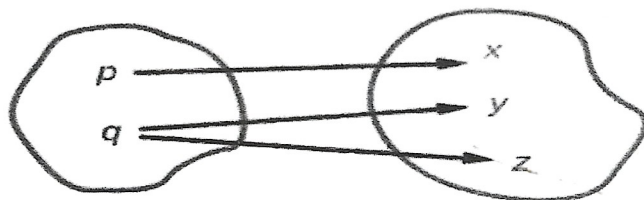
(x) Write down the constraints for the LPP below.



(xi) Examine graphically whether the function shown below is injective.

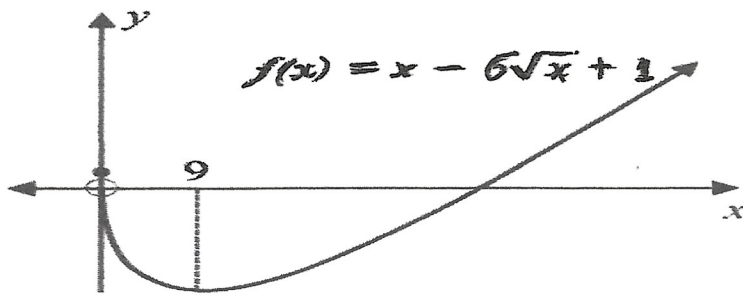


(xii) Write down the number of relations possible ?



(xiii) Given $y = f(x) = (3x + 5)(4x - 2)$, find $f^{-1}(y)$

(xiv) From the graph of the following function, find the image of 9.



(xv)

Assertion (A): Matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and matrix $B = \begin{bmatrix} 5 & -2 & 4 \\ 6 & 3 & -7 \end{bmatrix}$ then order of matrix AB is 3×3

Reason (R): If the order of matrix A is $m \times n$ and order of matrix B is $n \times p$ then order of matrix $A \times B$ is $m \times p$.

- (a) A and R are both true and R is the correct explanation of A
- (b) A and R are both true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

(xvi) A bag contains 8 red and 5 white balls. Two successive draws of 3 balls are made with replacement. Find the probability that the first drawing will give 3 white balls and the second 3 red balls.

(xvii) If $\cos^{-1}(x/x+1) = \sin^{-1}(B)$, find B in terms of x .

(xviii) If $A(2, -1, 5)$ & $B(3, 0, 1)$ are two points, find a vector of magnitude 10 units along AB vector.

(xix) Is the following matrix invertible? Justify.

$$C = \begin{bmatrix} -1 & -2 \\ 6 & 3 \end{bmatrix}$$

(xx) An objective function $Z = ax + by$ is maximum at points $(8, 10)$ and $(7, 12)$. If $a, b \geq 0$ and $ab = 72$, then the maximum value of the function is equal to?

Section B (14 Marks)

Question 2

Show that the relation defined on the set of Reals by $x \leq y^3$, is not reflexive or symmetric.

Question 3

Evaluate $\tan^{-1}(\tan(70\pi/3))$

Question 4

If A & B are symmetric matrices, show that $A + B$ is symmetric.

Question 5

If $\vec{a} = 5\hat{i} - \hat{j} + 7\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \lambda\hat{k}$, find λ such that $\vec{a} - \vec{b}$ and $\vec{a} + \vec{b}$ are perpendicular to each other.

Question 6

Find the angle between two vectors if they have same length $\sqrt{2}$ and their scalar product is -1 .

Question 7

$$\text{If } \begin{bmatrix} x & 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 2 \\ 1 & 0 & 2 \\ 0 & 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ -1 \end{bmatrix} = 0, \text{ find } x.$$

Question 8

A bag contains 4 red, 3 blue and 2 white balls. If three balls are drawn one by one (without replacement), what is the probability that all three balls are red?

Section C(21 Marks)

Question 9

A bag contains 10 white and 6 black balls. 4 balls are successively drawn out without replacement. is the probability that they are alternately of different colours?

Question 10

Given $f(x) = x/x+1$, find $f(f(f(x)))$

Question 11

If $(\vec{a} - \vec{b}) \cdot (\vec{a} + \vec{b}) = 8$ and $|\vec{a}| = 8|\vec{b}|$, find $|\vec{a}|$ and $|\vec{b}|$.

Question 12

$$\begin{bmatrix} x+1 & -3 & 4 \\ -5 & x+2 & 2 \\ 4 & 1 & x-6 \end{bmatrix} \text{ is singular.}$$

Find the values of x . Is it possible to find the inverse of this matrix?

Question 13

$$\begin{bmatrix} 2 & 3 & 4 \\ 5 & 7 & 9 \\ -2 & 1 & 1 \end{bmatrix}$$

Express as the sum of a symmetric and skew symmetric matrix.

Question 14

$$\frac{2}{x} + \frac{3}{y} = 2$$

$$\frac{5}{x} + \frac{8}{y} = \frac{31}{6}$$

Solve the above equations using Martin's Rule

Question 15

A and B throw a coin alternately till one of them gets a head and wins the game. Find their respective probabilities of winning. (They follow the same order)

SECTION D(25 Marks)

Question 16

a) The sum of three numbers is 20. If we multiply the first number by 2 and add the second number to the result and subtract the third number, we get 23. By adding second and third numbers to three times the first number, we get 46. Represent the above problem algebraically and use Martin's rule to find the numbers from these equations.

OR

b) Solve by Matrix Method

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$$

$$\frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1$$

$$\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2.$$

Question 17

a) Study the following distribution and answer the relevant questions :

$$P(X = x) = \begin{cases} 0.1 & , \text{if } x = 0 \\ kx & , \text{if } x = 1 \text{ or } 2 \\ k(5-x) & , \text{if } x = 3 \text{ or } 4 \\ 0 & , \text{otherwise} \end{cases}$$

where k is a constant.

(i) Find the value of k .

(ii) What is the probability that you study atleast two hours? Exactly two hours? Atmost two hours?

OR

b) Two cards are drawn simultaneously from a pack of cards. Find the probability distribution of the number of aces. Also find the expectation.

Question 18

Kiran plays a game of throwing a fair die 3 times, but quits as soon as a six is obtained.

Kiran gets 2 points for throwing a six, but loses 1 point for not throwing a six.

Write down the distribution of the number of sixes and find the expectation and also the probability of getting at least one 6.

Question 19

Find the product of matrices A and B where :

$$A = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}.$$

Hence, solve the following system of equations by matrix method :

$$x + y + 2z = 1, \quad 3x + 2y + z = 7, \quad 2x + y + 3z = 2$$

Question 20

a)

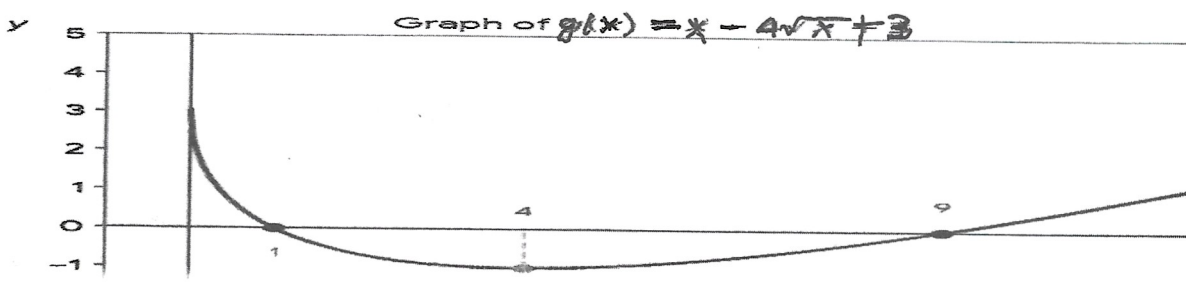
Based on the following graph answer the following :

i) State the domain and range

ii) Is the function invertible in $[0, 4]$. Justify.

iii) Find the x-intercepts

iv) Is the function invertible in $[9, \infty)$



OR

b) There are three identical cards, except that both sides of the first card are coloured red, both sides of the second card are coloured blue, and the third card has one red side and one blue side. One card is randomly selected from these three cards and placed face down. If the visible side of the card is red, what is the probability that the other side is blue, using Baye's theorem?

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